

CS343 Operating Systems

Lecture 1

Operating Systems – Classifications and Service



John Jose

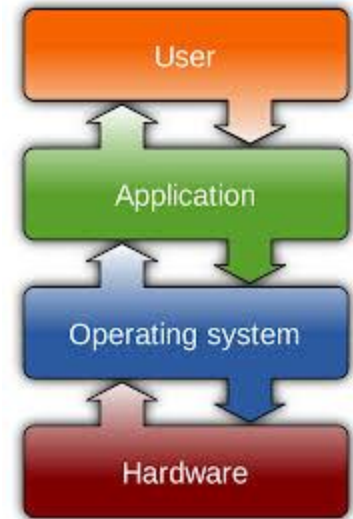
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What is an Operating System?

- ❖ A program that acts as an intermediary between a user of a computer and the computer hardware
- ❖ Operating system goals:
 - ❖ Execute user programs on hardware
 - ❖ Make the computer system convenient to use
 - ❖ Use the computer hardware in an efficient manner



Types of Operating Systems

- ❖ Most systems use a single general-purpose/special purpose processor
- ❖ There are several architectures which all require a different OS:
 - ❖ Multiprogramming Systems
 - ❖ Desktop PCs
 - ❖ Parallel Systems
 - ❖ Clustered Systems
 - ❖ Real-time Systems
 - ❖ Embedded Systems

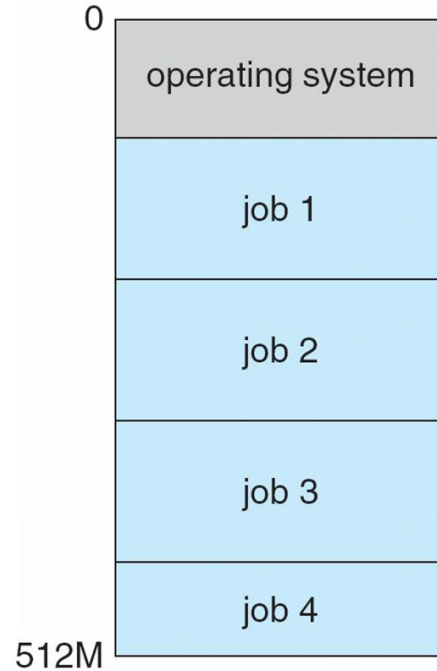
Multiprogramming Systems

- ❖ **Multiprogramming** (**Batch system**) needed for efficiency
 - ❖ Single user cannot keep CPU and I/O devices busy at all times
 - ❖ Multiprogramming organizes jobs (code and data) so CPU always has one to execute
 - ❖ A subset of total jobs in system is kept in memory
 - ❖ One job selected and run via **job scheduling**
 - ❖ When it has to wait (for I/O), OS switches to another job

Timesharing Systems

- ❖ **Timesharing** (**multitasking**) is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive** computing
 - ❖ Multiple programs executing in memory
 - ❖ If several jobs ready to run at the same time ⇒ **CPU scheduling**

Memory Layout for Multiprogrammed System



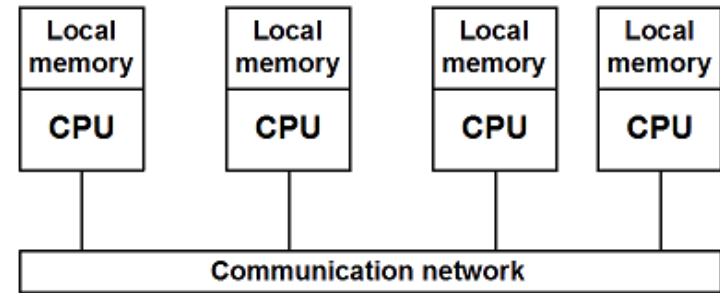
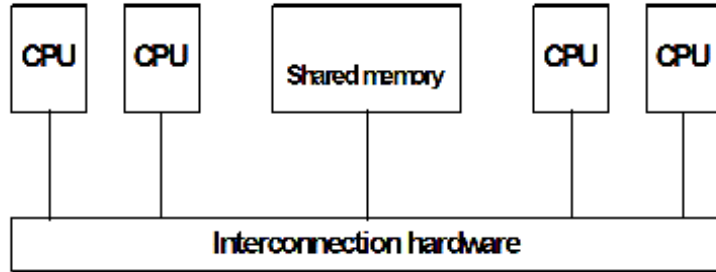
Desktop PCs

- ❖ Personal Computers – computer system dedicated to a single user.
- ❖ I/O devices – keyboards, mice, display screens, small printers.
- ❖ User convenience and responsiveness.
- ❖ Mostly single user - do not need advanced CPU utilization or protection features.
- ❖ May run several different types of operating systems (Windows, MacOS, UNIX, Linux)



Parallel Systems

- ❖ Multiprocessor systems with more than one CPU in close communication.
- ❖ **Tightly coupled system** – processors share memory and the internal clock; communication usually takes place through the shared memory.
- ❖ **Loosely coupled system** – multiple processors/computers with its own memory connected together for efficiency and throughput.



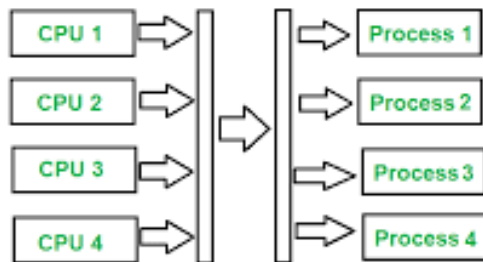
Multiprocessor/Parallel Systems

- ❖ **Multiprocessors** systems growing in use and importance
 - ❖ **Increased throughput**
 - ❖ **Economy of scale**
 - ❖ **Increased reliability** – graceful degradation or fault tolerance
 - ❖ **Asymmetric Multiprocessing**
 - ❖ **Symmetric Multiprocessing**

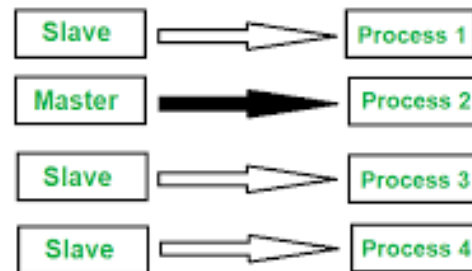
Multiprocessor Systems

- ❖ **Asymmetric Multiprocessing**
- ❖ Each processor is assigned a specific task; master processor schedules and farms work to slave processors.
- ❖ More common in extremely large systems like mainframes with hundreds of processors.

Symmetric Multiprocessing



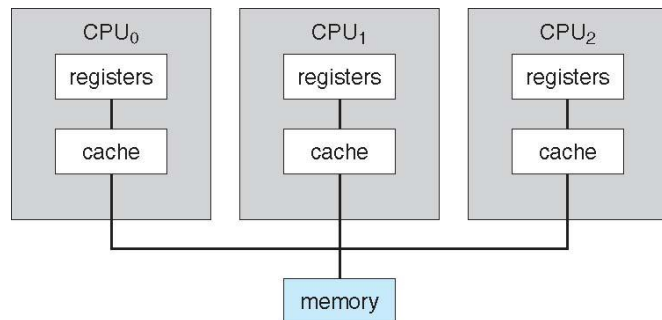
Asymmetric Multiprocessing



Multiprocessor Systems

❖ Symmetric Multiprocessing

- ❖ Each processor runs an identical copy of the operating system.
- ❖ The OS code is usually shared.
- ❖ Many processes can run at once without performance deterioration.
- ❖ Most modern operating systems have SMP support.
- ❖ OS has to cater for protection of data.

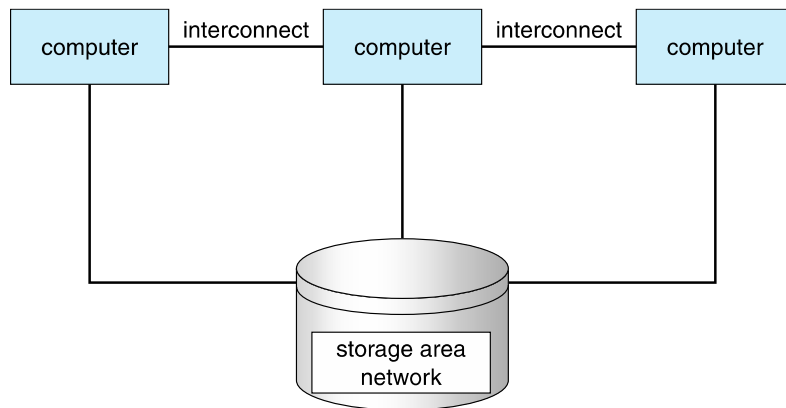


Clustered Systems

- ❖ Like multiprocessor systems working together
 - ❖ Usually sharing storage via a **storage-area network (SAN)**
 - ❖ Provides a **high-availability** service which survives failures
 - ❖ **Asymmetric clustering** has one machine in hot-standby mode
 - ❖ **Symmetric clustering** has multiple nodes running applications, monitoring each other

Clustered Systems

- ❖ Some clusters are for **high-performance computing (HPC)**
 - ❖ Applications must be written to use **parallelization**
- ❖ Some have **distributed lock manager (DLM)** to avoid conflicting operations



Real-Time Systems

- ❖ Often used as a control device in a dedicated application such as controlling scientific experiments, medical imaging systems, industrial control systems, and some display systems.
- ❖ Well-defined fixed-time constraints.
- ❖ Real-Time systems may have either hard or soft real-time.

Embedded Systems

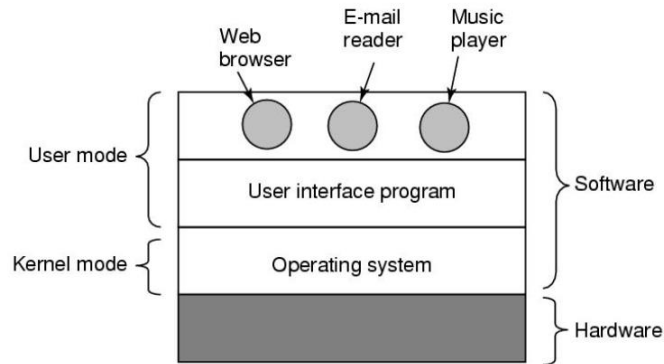
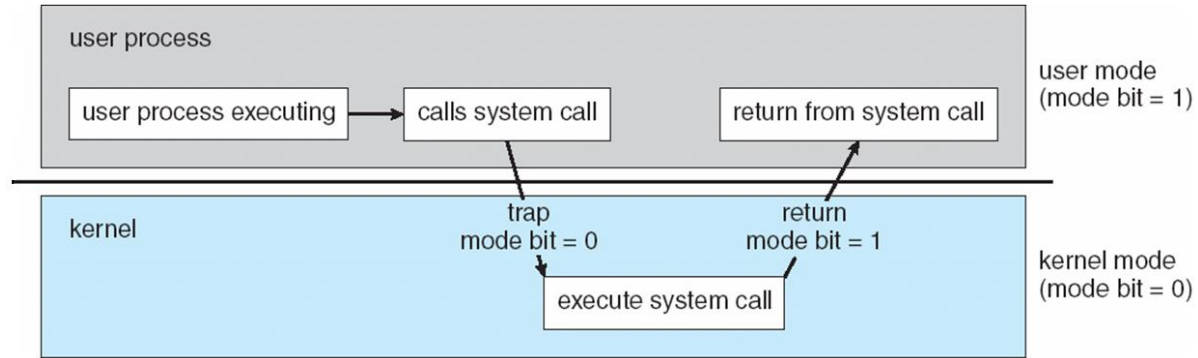
- ❖ Personal Digital Assistants (PDAs)
- ❖ Smart telephones
- ❖ Issues:
 - ❖ Limited memory, Slow processors, Small display screens.
 - ❖ Emphasis is on I/O operations.
 - ❖ Limited memory management and protection



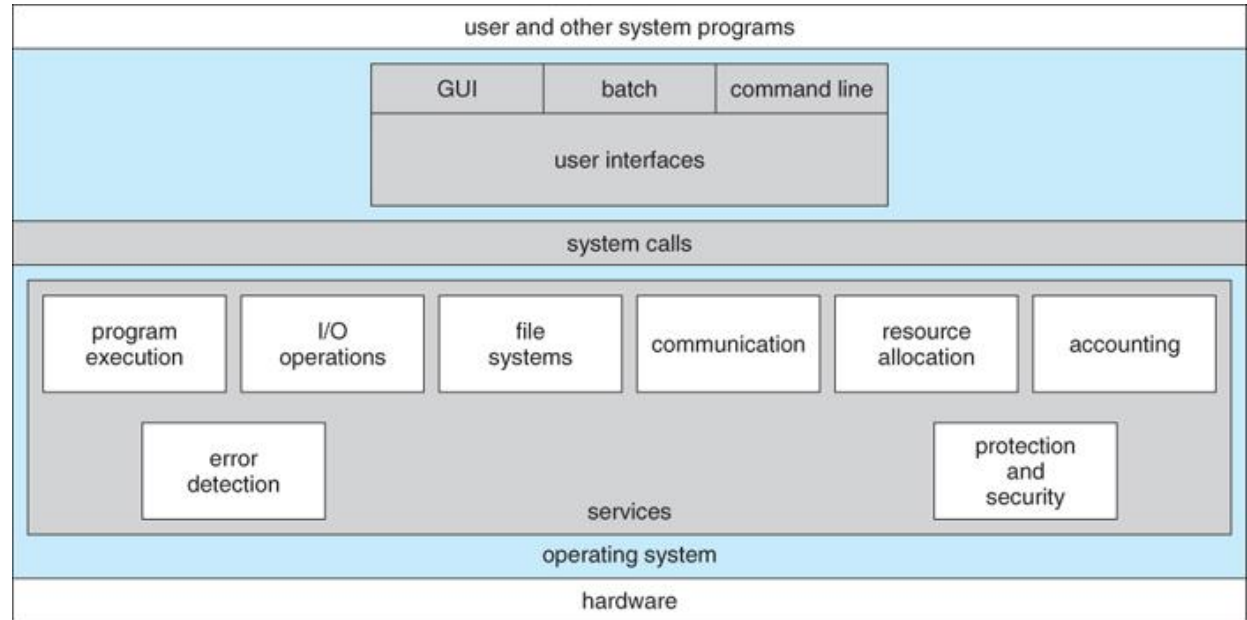
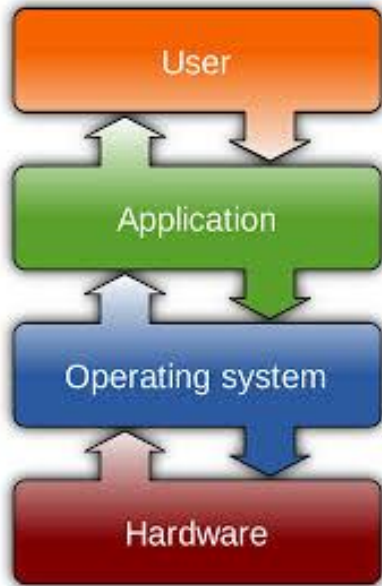
Operating System in Dual Mode

- ❖ **Dual-mode** operation allows OS to protect itself and other system components
 - ❖ **User mode** and **kernel mode**
 - ❖ **Mode bit** provided by hardware
 - ❖ Provides ability to distinguish when system is running user code or kernel code
 - ❖ Some instructions designated as **privileged**, only executable in kernel mode
 - ❖ System call changes mode to kernel, return from call resets it to user

Operating System in Dual Mode



Operating System Services



Operating System Services

- ❖ The OS structure is divided into many sub-components.
 - ❖ Process Execution
 - ❖ Process Management
 - ❖ Memory Management
 - ❖ File Management
 - ❖ Storage Management
 - ❖ I/O Sub-system Management
 - ❖ Protection and Security
 - ❖ User Interface

Process Execution

- ❖ A process is a **program in execution**
- ❖ It is a unit of work within the system
- ❖ Program is a **passive entity**, process is an **active entity**.
- ❖ OS must be able to load a program into memory, run that program, and end execution
- ❖ **Assign resources** like CPU, memory, I/O, files, data to accomplish its task
- ❖ Process termination requires **reclaim of any reusable resources**

Process Management

- ❖ Creating and deleting both user and system processes
- ❖ Suspending and resuming processes
- ❖ Providing mechanisms for process synchronization
- ❖ Providing mechanisms for process communication
- ❖ Providing mechanisms for deadlock handling

Memory Management

- ❖ To execute a program all (or part) of the instructions must be in memory
- ❖ All (or part) of the data that is needed by the program must be in memory
- ❖ Memory management determines what is in memory and when
- ❖ Keeping track of which parts of memory are currently being used and by whom
- ❖ Deciding which processes and data to move into and out of memory
- ❖ Allocating and deallocating memory space as needed

File Management

- ❖ OS provides uniform, logical view of information storage
- ❖ Abstracts physical properties of storage to logical storage unit - file
 - ❖ Files are usually organized into directories
 - ❖ OS determines access control on files/directories that determine who can access what
- ❖ File-System management include
 - ❖ Creating and deleting files and directories
 - ❖ Primitives to manipulate files and directories
 - ❖ Mapping files onto secondary storage
 - ❖ Backup files onto stable (non-volatile) storage media

Storage Management

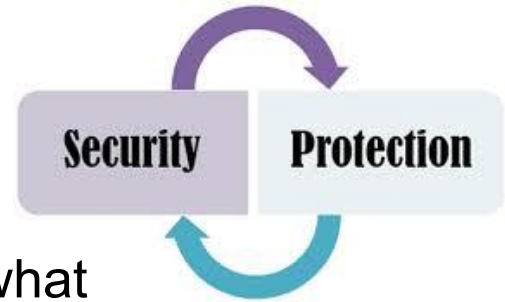
- ❖ Mass storage devices (disks/ tape drives) store data that does not fit in main memory or data that must be kept for a long period of time
- ❖ Devices vary in access speed, capacity, data-transfer rate, access method
- ❖ OS activities in disk management includes
 - ❖ Free-space management
 - ❖ Storage allocation
 - ❖ Disk scheduling

I/O Subsystem Management

- ❖ OS hides peculiarities of hardware devices from the user
- ❖ I/O subsystem responsible for
 - ❖ **Buffering** (storing data temporarily while it is being transferred)
 - ❖ **Caching** (storing parts of data in faster storage for performance)
 - ❖ **Spooling** (the overlapping of output of one job with input of other jobs)
 - ❖ **Providing** device-driver interface

Protection and Security

- ❖ **Protection** – any mechanism for controlling access of processes or users to resources defined by the OS
- ❖ **Security** – defense of the system against internal and external attacks
 - ❖ Huge range, including denial-of-service, worms, viruses, identity theft, theft of service
- ❖ Distinguish among users, to determine who can do what
 - ❖ User identities (**user IDs**) and associated access controls on resources
 - ❖ Group identifier (**group ID**) associated access controls on resources
 - ❖ **Privilege escalation** to change to give more rights



User Interface

- ❖ Provides a **user-friendly platform** to initiate actions from user side.
- ❖ The UI primarily **receives command** from user and executes it
- ❖ Command-Line Interface (CLI) allows direct command entry
- ❖ User-friendly desktop Graphical User Interface (GUI)
 - ❖ Usually mouse, keyboard, and monitor used for giving inputs.
 - ❖ Icons represent files, programs, actions, etc
 - ❖ Various mouse buttons over objects in the interface cause various actions (provide information, options, execute function, open directory)
- ❖ Many systems (Microsoft, Apple-Mac OS, UNIX) now include both CLI and GUI interfaces



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